



**Autonomous Cyber Defense, Risk Intelligence & Pre-Breach
Simulation Platform**

COMPLETE SYSTEM DOCUMENTATION & OPERATIONAL MANUAL

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Status: CONFIDENTIAL

1. System Overview

ACDRIP+ is an advanced cybersecurity platform designed for autonomous threat detection, risk quantification, and attack path simulation. It integrates real-time network scanning with machine learning to provide financial risk impact assessments and MITRE ATT&CK; alignment.

2. Technology Stack

The platform is built using a modern, scalable stack designed for high-performance security operations.

Layer	Technology	Description
Backend	FastAPI (Python 3.10+)	High-performance asynchronous API framework
Frontend	HTML5 / Vanilla JS / CSS3	Ultra-fast, zero-dependency UI with Glassmorphism
Database	SQLite / SQLAlchemy	Relational storage with ORM abstraction
Security	JWT / Bcrypt	Industry-standard authentication and hashing
Scanning	Nmap / Python-Nmap	Network reconnaissance and service discovery
ML Engine	Scikit-Learn	Gradient Boosting & Random Forest for risk prediction
Visualization	Chart.js	Dynamic data visualization and metrics

3. Architecture Flow

ACDRIP+ follows a decoupled Client-Server architecture:

- Request Phase**: The Frontend sends JSON requests with JWT headers to the FastAPI backend.
- Auth Layer**: Middleware validates the token and identifies the user context.
- Core Services**: The request is routed to specific modules (Scanner, Simulation, Risk).
- Data Persistence**: Results are committed to the SQLite database via SQLAlchemy sessions.
- Response Phase**: Data is returned as structured JSON, which the JS app renders dynamically.

4. Database Management

The primary database is 'acdrip_plus.db' located in the backend directory. It can be accessed using standard SQLite clients.

```
Connection String: sqlite:///./acdrip_plus.db
```

Key Tables:

- users**: Stores identity and password hashes.

- ****scans****: Stores historical network scan results and risk levels.
- ****vulnerabilities****: Detailed CVE mapping for each scan.
- ****reports****: Metadata and paths for generated PDF artifacts.

5. Operational Usage

A. Network Scanner

Input a target IP or domain. The system performs service discovery and maps discovered versions to the CVE database.

B. Dark Web Hunter

Analyzes underground forums and breach databases for exposed credentials and asset mentions related to the target domain.

C. Quantum Threat Intel

Audits cryptographic algorithms (RSA, ECC) for susceptibility to Shor's and Grover's algorithms, providing a PQC (Post-Quantum Cryptography) migration score.

END OF DOCUMENTATION